

Quantum Computing, Quantum Machine Learning and Quantum Information Theories

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February 5, 2025

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Plans for the week of February 3-7: Entanglement, entropies and density matrices

1. Reminder and review of density matrices and measurements from last week
2. Discussion of entropies, classical information entropy (Shannon entropy) and von Neumann entropy
3. Single-qubit and two-qubit gates and codes. Link to exercises
4. Video of lecture to be added

Reminder on density matrices and traces

We have the spectral decomposition of a given operator $\mathbf{A}$ given by

$$\mathbf{A} = \sum_i \lambda_i |i\rangle\langle i|,$$

with the ONB $|i\rangle$ being eigenvectors of $\mathbf{A}$ and λ_i being the eigenvalues. Similarly, a operator which is a function of $\mathbf{A}$ is given by

$$f(\mathbf{A}) = \sum_i f(\lambda_i) |i\rangle\langle i|.$$

The trace of a product of matrices is cyclic, that is

$$\mathrm{tr}[\mathbf{ABC}] = \mathrm{tr}[\mathbf{BCA}] = \mathrm{tr}[\mathbf{CBA}],$$

and we have also

$$\mathrm{tr}[\mathbf{A}|\psi\rangle\langle\psi|] = \langle\psi|\mathbf{A}|\psi\rangle.$$

Definition of density matrix

Using the spectral decomposition we defined also the density matrix as

$$\rho = \sum_i p_i |i\rangle\langle i|,$$

where the probability p_i are the eigenvalues of the density linked with the ONB $|i\rangle$.

The trace of the density matrix is $\text{tr}\rho = 1$ and it is invariant under unitary transformations $|\psi'_i\rangle = \mathbf{U}|\psi_i\rangle$. The unitary transformation of the density matrix gives, with $\mathbf{U}^\dagger \mathbf{U} = \mathbf{U}^T \mathbf{U} = \mathbf{I}$,

$$\mathbf{U}\rho\mathbf{U}^\dagger = \sum_i p_i \mathbf{U}|\psi_i\rangle\langle\psi_i|\mathbf{U}^\dagger,$$

and with the unitary transformation it is easy to show that the trace of the transformed density matrix is equal to one,

$$\text{tr} [\mathbf{U}\rho\mathbf{U}^\dagger] = \text{tr} [\mathbf{U}\mathbf{U}^\dagger\rho] = 1.$$

First entanglement encounter, two qubit system

We define a system that can be thought of as composed of two subsystems A and B . Each subsystem has computational basis states

$$|0\rangle_{A,B} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T \quad |1\rangle_{A,B} = \begin{bmatrix} 0 & 1 \end{bmatrix}^T.$$

The subsystems could represent single particles or composite many-particle systems of a given symmetry.

Computational basis

This leads to the many-body computational basis states

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B = [1 \ 0 \ 0 \ 0]^T,$$

and

$$|01\rangle = |0\rangle_A \otimes |1\rangle_B = [0 \ 1 \ 0 \ 0]^T,$$

and

$$|10\rangle = |1\rangle_A \otimes |0\rangle_B = [0 \ 0 \ 1 \ 0]^T,$$

and finally

$$|11\rangle = |1\rangle_A \otimes |1\rangle_B = [0 \ 0 \ 0 \ 1]^T.$$

Bell states

The above computational basis states, which define an ONB, can in turn be used to define another ONB. As an example, consider the so-called Bell states

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} [|00\rangle + |11\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} [|00\rangle - |11\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix},$$

The next two

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} [|10\rangle + |01\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix},$$

and

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} [|10\rangle - |01\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}.$$

It is easy to convince oneself that these states also form an orthonormal basis.

Measurement

Measuring one of the qubits of one of the above Bell states, automatically determines, as we will see below, the state of the second qubit. To convince ourselves about this, let us assume we perform a measurement on the qubit in system A by introducing the projections with outcomes 0 or 1 as

$$P_0 = |0\rangle\langle 0|_A \otimes I_B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

for the projection of the $|0\rangle$ state in system A and similarly

$$P_1 = |1\rangle\langle 1|_A \otimes I_B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

for the projection of the $|1\rangle$ state in system A .

Probability of outcome

We can then calculate the probability for the various outcomes by computing for example the probability for measuring qubit 0

$$\langle \Phi^+ | \mathbf{P}_0 | \Phi^+ \rangle = \frac{1}{2} [\langle 00 | + \langle 11 |] \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} [|00\rangle + |11\rangle] = \frac{1}{2}.$$

Similarly, we obtain

$$\langle \Phi^+ | \mathbf{P}_1 | \Phi^+ \rangle = \frac{1}{2} [\langle 00 | + \langle 11 |] \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} [|00\rangle + |11\rangle] = \frac{1}{2}.$$

States after measurement

After the above measurements the system is in the states

$$|\Phi'_0\rangle = \sqrt{2} [|0\rangle\langle 0|_A \otimes I_B] |\Phi^+\rangle = |00\rangle,$$

and

$$|\Phi'_1\rangle = \sqrt{2} [|1\rangle\langle 1|_A \otimes I_B] |\Phi^+\rangle = |11\rangle.$$

We see from the last two equations that the state of the second qubit is determined even though the measurement has only taken place locally on system A .

Other states

If we on the other hand consider a state like

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B,$$

this is a pure **product** state of the single-qubit, or single-particle states, of two qubits (particles) in system A and system B , respectively. We call such a state for a **pure product state**.

Quantum states that cannot be written as a mixture of other states are called pure quantum states or just product states, while all other states are called mixed quantum states.

More on Bell states

A state like one of the Bell states (where we introduce the subscript AB to indicate that the state is composed of single states from two subsystem)

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} [|00\rangle_{AB} + |11\rangle_{AB}],$$

is on the other hand a mixed state and we cannot determine whether system A is in a state 0 or 1. The above state is a superposition of the states $|00\rangle_{AB}$ and $|11\rangle_{AB}$ and it is not possible to determine individual states of systems A and B , respectively.

Entanglement

We say that the state is entangled. This yields the following definition of entangled states: a pure bipartite state $|\psi\rangle_{AB}$ is entangled if it cannot be written as a product state $|\psi\rangle_A \otimes |\phi\rangle_B$ for any choice of the states $|\psi\rangle_A$ and $|\phi\rangle_B$. Otherwise we say the state is separable.

Examples of entanglement

As an example, consider an ansatz for the ground state of the helium atom with two electrons in the lowest $1s$ state (hydrogen-like orbits) and with spin $s = 1/2$ and spin projections $m_s = -1/2$ and $m_s = 1/2$. The two single-particle states are given by the tensor products of their spatial $1s$ single-particle states $|\phi_{1s}\rangle$ and their spin up or spin down spinors $|\xi_{sm_s}\rangle$. The ansatz for the ground state is given by a Slater determinant with total orbital momentum $L = l_1 + l_2 = 0$ and total spin $S = s_1 + s_2 = 0$, normally labeled as a spin-singlet state.

Ground state of helium

This ansatz for the ground state is then written as, using the compact notations

$$|\psi_i\rangle = |\phi_{1s}\rangle_i \otimes |\xi\rangle_{s_i m_{s_i}} = |1s, s, m_s\rangle_i,$$

with i being electron 1 or 2, and the tensor product of the two single-electron states as

$|1s, s, m_s\rangle_1 |1s, s, m_s\rangle_2 = |1s, s, m_s\rangle_1 \otimes |1s, s, m_s\rangle_2$, we arrive at

$$\Psi(\mathbf{r}_1, \mathbf{r}_2; s_1, s_2) = \frac{1}{\sqrt{2}} [|1s, 1/2, 1/2\rangle_1 |1s, 1/2, -1/2\rangle_2 - |1s, 1/2, -1/2\rangle_1 |1s, 1/2, 1/2\rangle_2]$$

This is also an example of a state which cannot be written out as a pure state. We call this for an entangled state as well.

Maximally entangled

A so-called maximally entangled state for a bipartite system has equal probability amplitudes

$$|\Psi\rangle = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |ii\rangle.$$

We call a bipartite state composed of systems A and B (these systems can be single-particle systems, or single-qubit systems representing low-lying states of complicated many-body systems) for separable if its density matrix ρ_{AB} can be written out as the tensor product of the individual density matrices ρ_A and ρ_B , that is we have for a given probability distribution p_i

$$\rho_{AB} = \sum_i p_i \rho_A(i) \otimes \rho_B(i).$$

Entropies and density matrices

Note: more details on whiteboard. This material will be added later

Shannon information entropy

We start our discussions with the classical information entropy, or just Shannon entropy, before we move over to a quantum mechanical way to define the entropy based on the density matrices discussed earlier.

We define a set of random variables $X = \{x_0, x_1, \dots, x_{n-1}\}$ with probability for an outcome $x \in X$ given by $p_X(x)$, the information entropy is defined as

$$S = - \sum_{x \in X} p_X(x) \log_2 p_X(x).$$

Von Neumann entropy

$$S = -\mathrm{Tr}[\rho \log_2 \rho].$$

Two-qubit system and calculation of density matrices and exercise

This part is best seen using the jupyter-notebook.

This system can be thought of as composed of two subsystems A and B . Each subsystem has computational basis states

$$|0\rangle_{A,B} = [1 \ 0]^T \quad |1\rangle_{A,B} = [0 \ 1]^T.$$

The subsystems could represent single particles or composite many-particle systems of a given symmetry. This leads to the many-body computational basis states

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